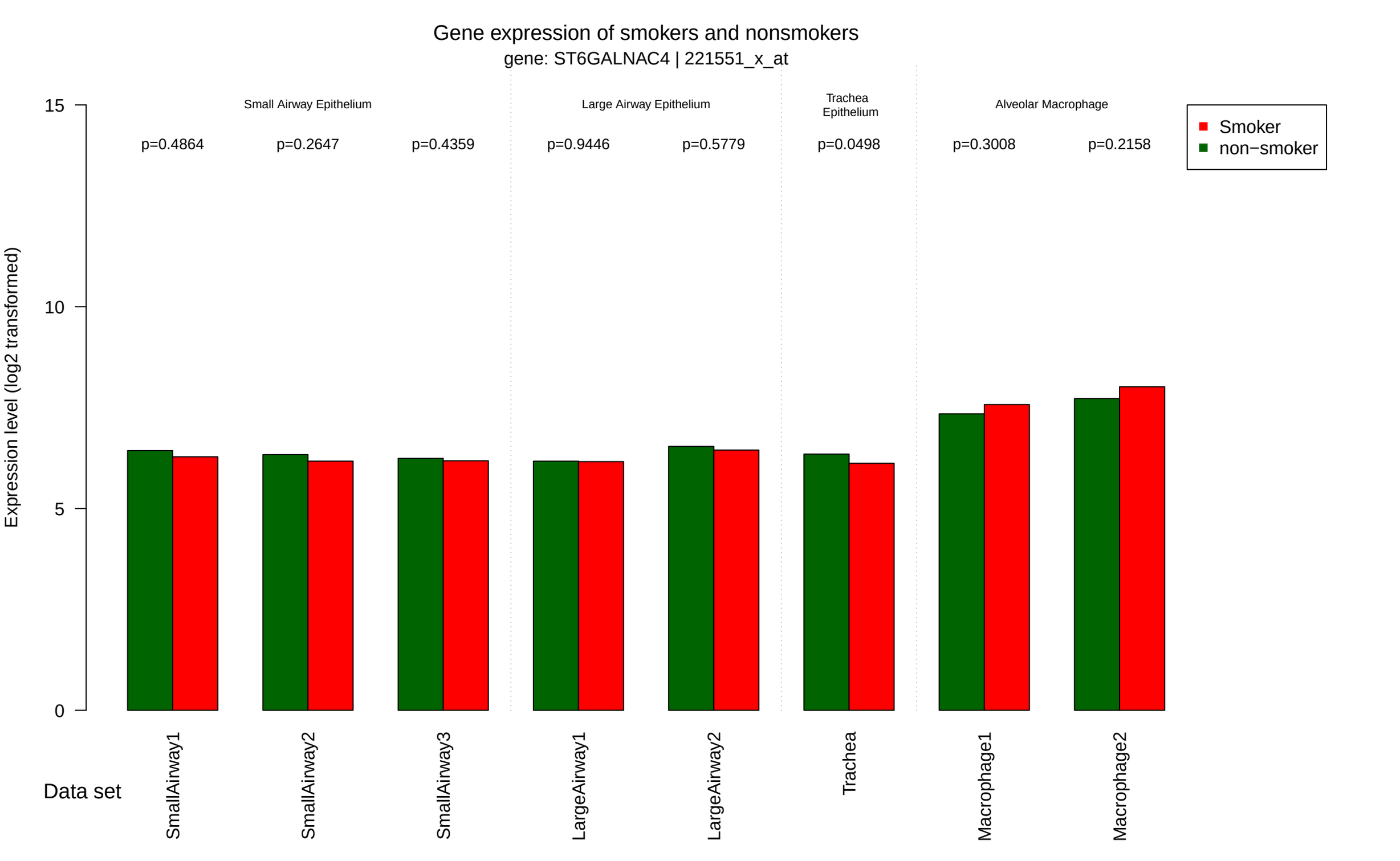
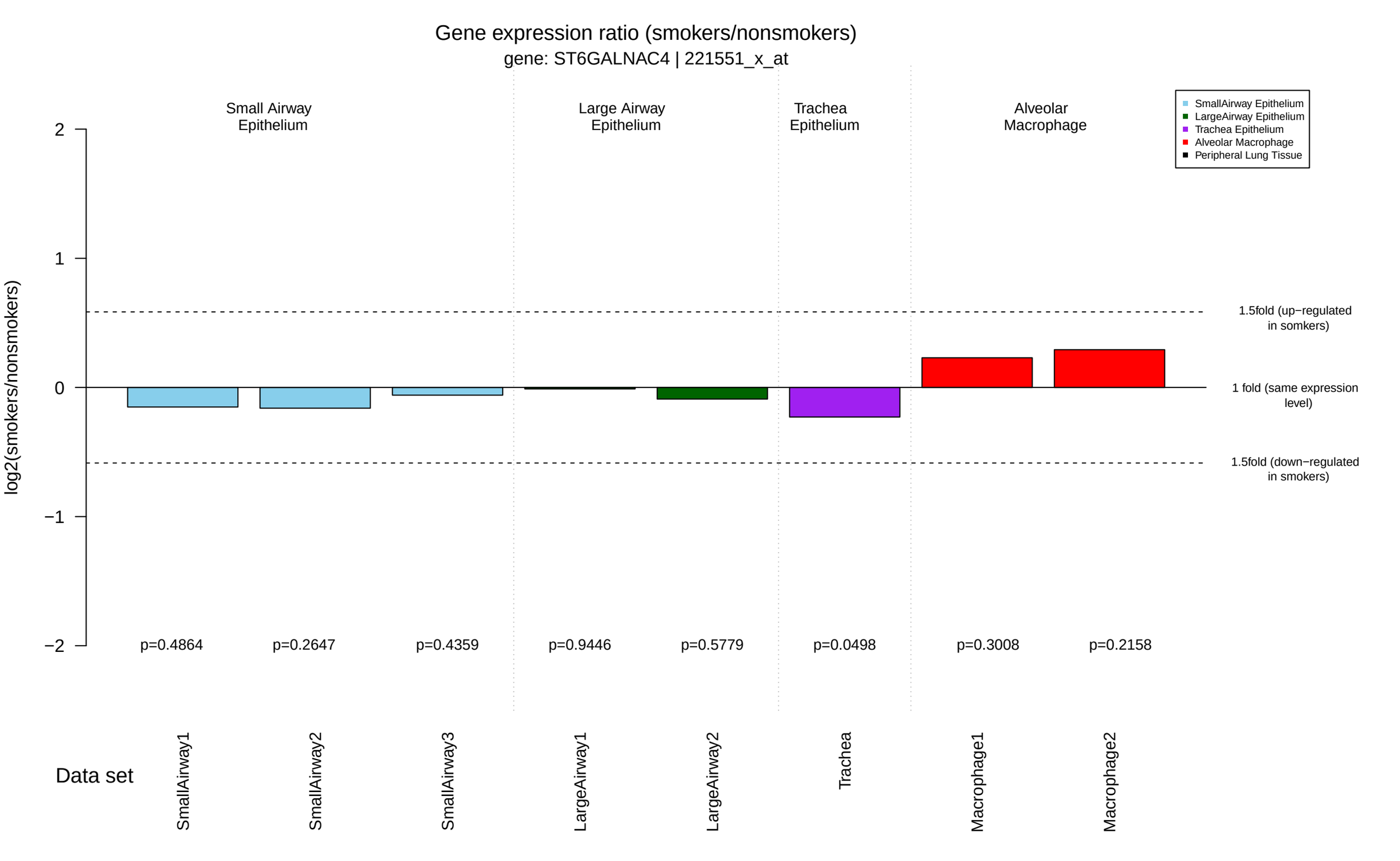
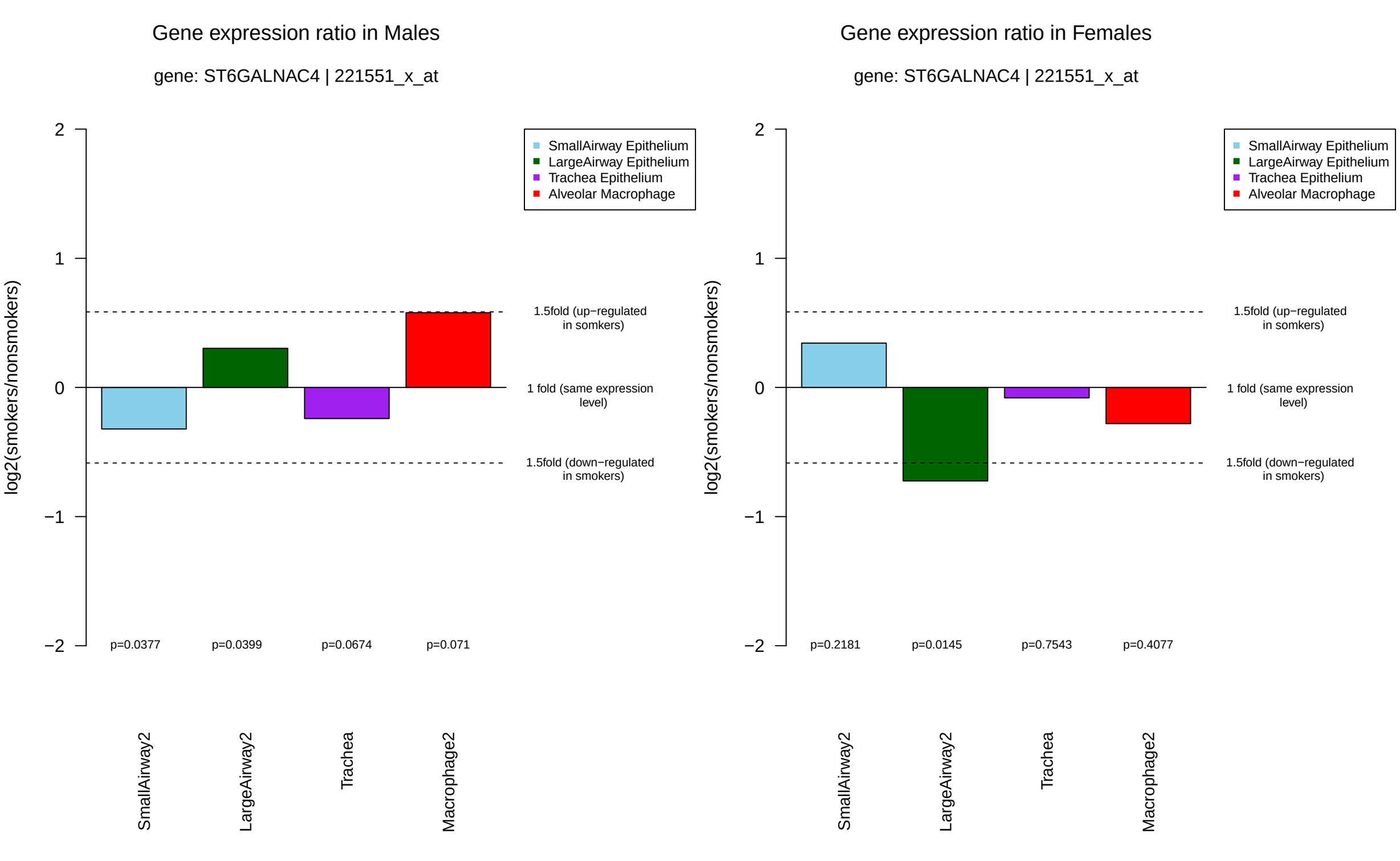
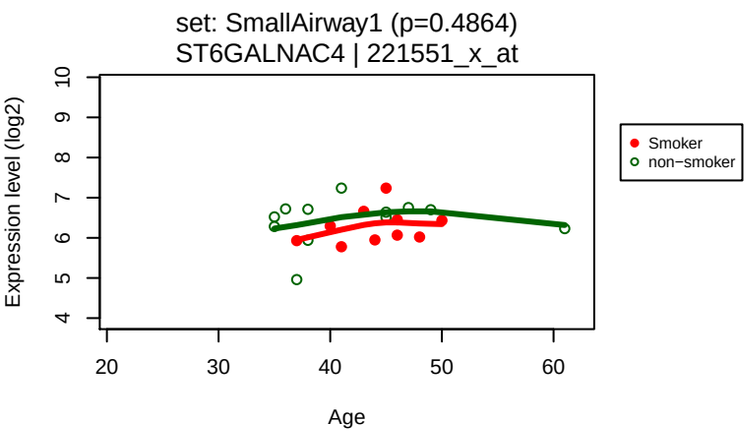
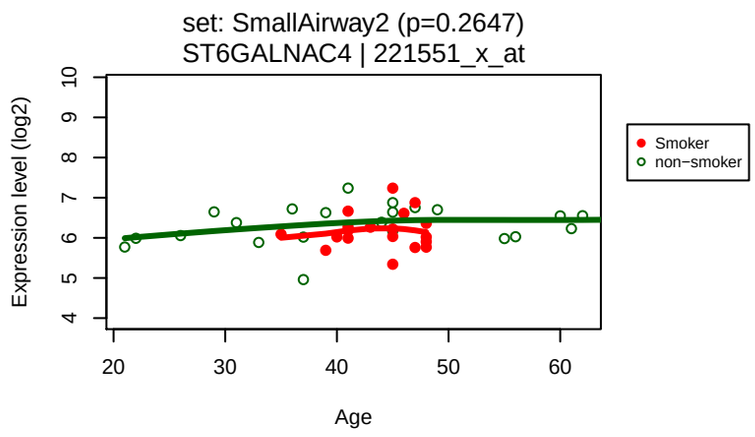
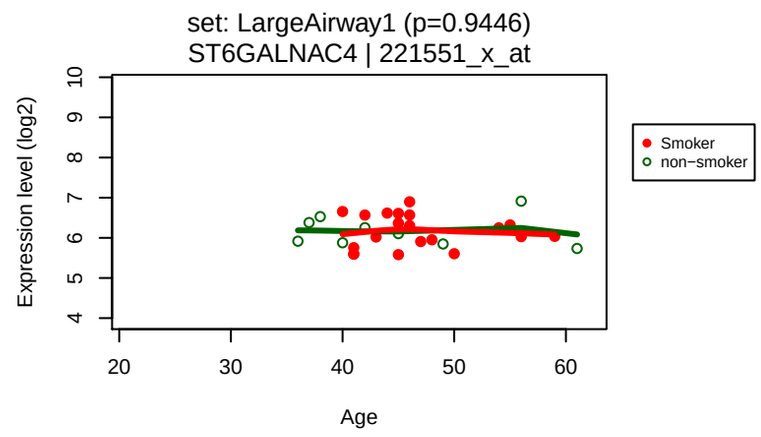

Gene expression of smokers and nonsmokers
gene: ST6GALNAC4 | 221551_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4864
p=0.2647
p=0.4359
p=0.9446
p=0.5779
p=0.0498
p=0.3008
p=0.2158
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

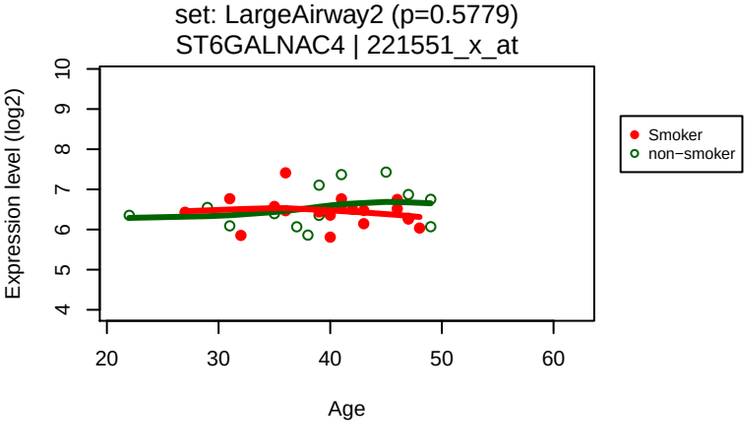

Gene expression ratio (smokers/nonsmokers)
gene: ST6GALNAC4 | 221551_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4864
p=0.2647
p=0.4359
p=0.9446
p=0.5779
p=0.0498
p=0.3008
p=0.2158
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

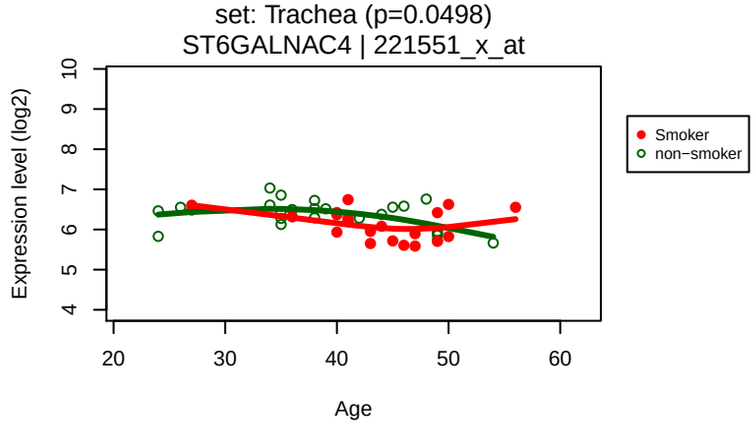

Gene expression ratio in Males
gene: ST6GALNAC4 | 221551_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0377
p=0.0399
p=0.0674
p=0.071
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST6GALNAC4 | 221551_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2181
p=0.0145
p=0.7543
p=0.4077
SmallAirway2
LargeAirway2
Trachea
Macrophage2

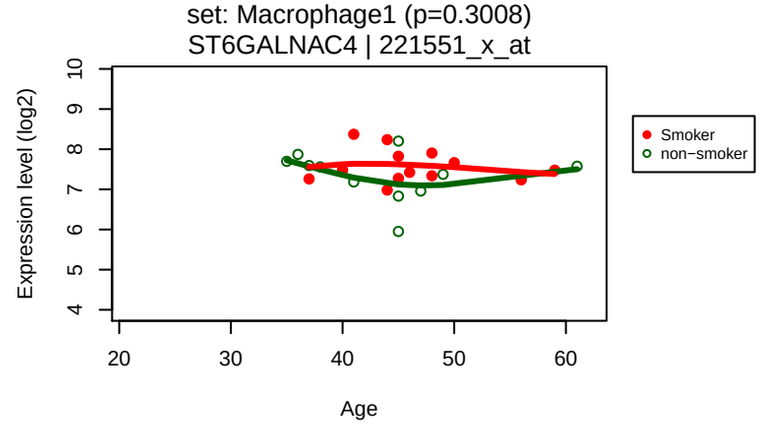

set: SmallAirway1 (p=0.4864)
ST6GALNAC4 | 221551_x_at
Expression level (log2)
Age
Smoker
non−smoker

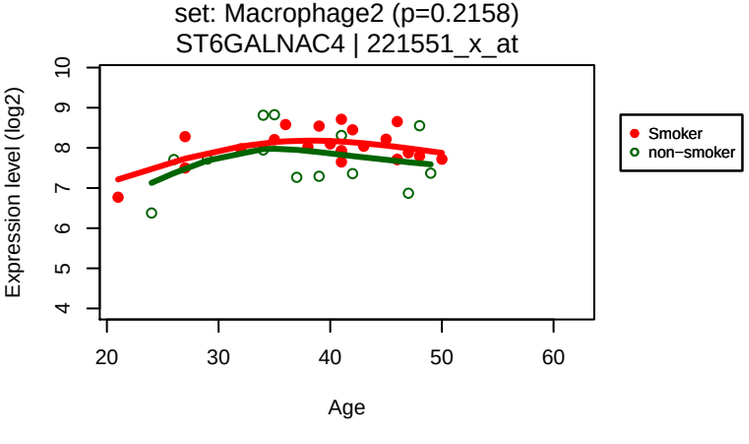

set: SmallAirway2 (p=0.2647)
ST6GALNAC4 | 221551_x_at
Expression level (log2)
Age
Smoker
non−smoker

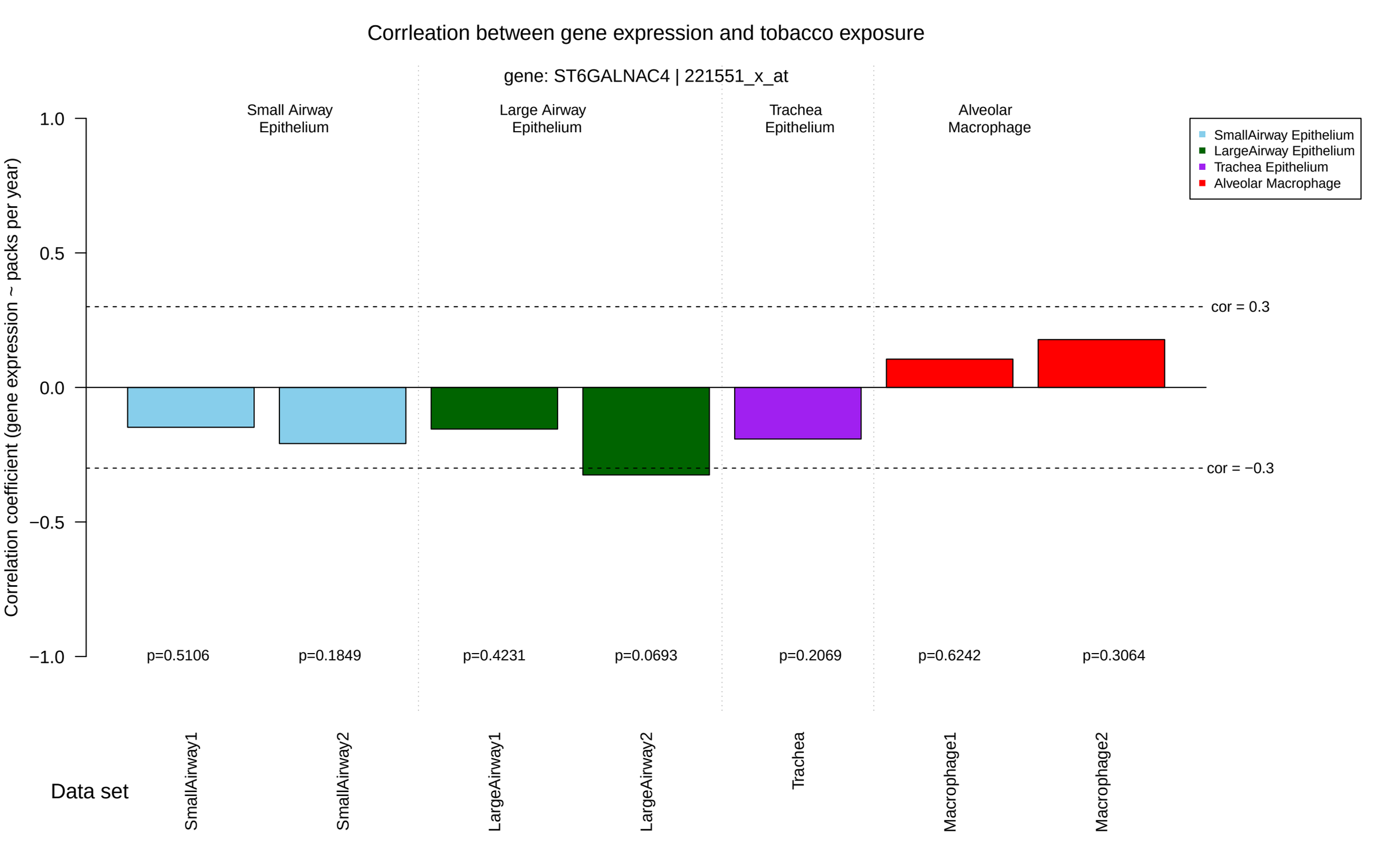

set: LargeAirway1 (p=0.9446)
ST6GALNAC4 | 221551_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5779)
ST6GALNAC4 | 221551_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0498)
ST6GALNAC4 | 221551_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3008)
ST6GALNAC4 | 221551_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2158)
ST6GALNAC4 | 221551_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST6GALNAC4 | 221551_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5106
p=0.1849
p=0.4231
p=0.0693
p=0.2069
p=0.6242
p=0.3064
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2